

VIKAS V

AI & Machine Learning Engineer | Automation | Software Development

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Portfolio: <https://portfolio2-olive-pi.vercel.app/>

Professional Summary

Computer Science Engineering graduate with strong focus on **Artificial Intelligence, Machine Learning, and Intelligent System Development**, with hands-on experience in building **real-time AI-driven applications** and **automation workflows**. Developed end-to-end solutions using **ANN, SVM, YOLO, Reinforcement Learning (PPO), FastAPI, Streamlit, and Godot**, including an **AI-based Smart Agriculture & Irrigation Recommendation System** integrating **live sensor monitoring, weather analysis, AI decision systems, and interactive visualization**. Built automation pipelines using **n8n** and **OpenAI APIs** for intelligent data processing and decision support. Published **IEEE research** in healthcare AI applying **machine learning** and **computer vision** techniques.

Research & Publication

IEEE Conference Paper – Early Diagnosis and Classification of Fetal Health and Brain Abnormality using Cardiotocography Dataset

Author

- Developed machine learning models using ANN, SVM, and YOLO for medical diagnosis
- Applied image processing techniques on clinical datasets for feature extraction
- Evaluated model performance for classification and detection tasks
- Demonstrated AI-driven healthcare system for early-stage diagnosis

Internship Experience

CYBERLANCERS

FEB 2026 – MAY 2026

AI Transformational Internship

- Working on AI-based applications involving machine learning and automation systems
- Developing data-driven models for real-world problem-solving scenarios
- Designing and implementing automation workflows using APIs and n8n
- Integrating AI models into practical systems for decision-making tasks
- Collaborating on project development, testing, and iterative improvements
- Exploring optimization techniques for improving model performance and system efficiency

CODTECH

Python Programming Intern

- Developed Python scripts for automation, data handling, and basic analytics
- Applied logical problem-solving techniques to build efficient solutions
- Worked with structured data and implemented reusable code modules
- Gained hands-on experience in debugging, testing, and optimizing Python programs

BHARATH INTERNS

Web Development Intern

- Built responsive frontend applications including Netflix clone and weather system
- Improved UI/UX and ensured cross-device compatibility
- Strengthened understanding of frontend architecture and component-based design

Projects

AI-Based Smart Agriculture & Irrigation System

- Developed an AI-powered agriculture platform for real-time crop monitoring and intelligent irrigation recommendation using environmental and soil sensor parameters
- Integrated FastAPI backend, OpenWeather API, and multi-zone live sensor simulation for dynamic agricultural data processing and monitoring
- Built interactive 3D crop visualization in Godot with health-based plant states and synchronized live backend updates
- Designed a Streamlit analytics dashboard featuring irrigation trends, nutrient analysis, crop health visualization, and AI-generated decision explanations

AI Irrigation Recommendation System

- Developed AI-based irrigation system to optimize water usage under varying conditions
- Built simulation environment using environmental parameters (soil moisture, temperature, humidity, rainfall)
- Implemented Reinforcement Learning using Gymnasium and DQN
- Trained the model over **100,000 iterations**, improving decision stability and policy learning
- Structured system for extensibility into fertilizer and yield prediction modules

AI Web Traffic Analyzer

- Developed AI system to analyse and monitor web traffic patterns and trends using n8n
- Built automation workflows using n8n for data collection, transformation, and routing
- Integrated OpenAI APIs for intelligent insights, anomaly detection, and summarization
- Designed reusable pipelines for continuous monitoring and analysis

n8n Automation Workflows

- Designed and implemented **3+ automation workflows** for real-time task execution
- Built event-driven pipelines using triggers, scheduling, and webhooks
- Automated API integrations and data routing across multiple services
- Improved workflow efficiency by reducing manual intervention

Fetal Health Diagnosis System (IEEE Project)

- Developed machine learning-based healthcare system for early diagnosis
- Applied ANN, SVM, and YOLO models for classification and detection
- Performed preprocessing and feature extraction on medical datasets
- Published IEEE research paper based on system performance

Web Development Projects

Full Stack Applications

Cafe Management System

- Designed and implemented a full-stack Cafe Management System with database integration for efficient order and user management

E-Learning Platform

- Built an E-Learning Platform with authentication, course management, and interactive user modules for seamless learning experience

Frontend Applications

Weather Forecast Web Application

- Developed a Weather Forecast Web Application with real-time API integration and dynamic user interface updates

Netflix Clone

- Built a responsive Netflix Clone replicating modern streaming platform UI with structured layouts and reusable components

Workshops

Web Development Workshop

- Learned HTML, CSS, and JavaScript fundamentals through hands-on implementation
- Developed **Cafe Management Web Application (Frontend Only)**
- Designed responsive UI and implemented interactive client-side functionality

Certification

- NPTEL – Big Data Analytics
- NPTEL – Java Programming
- Udemy – Java Programming Masterclass
- Celonis – Process Mining
- Zenith Adept – Coding in C++

Technical Skill

- **Languages/Frameworks:** C, C++, Java, Python, JavaScript, HTML, CSS, MySQL
- **Machine Learning & AI:** ANN, SVM, YOLO, Reinforcement Learning (DQN), PyTorch (basic)
- **Tools & Platforms:** n8n, OpenAI API, Prometheus
- **Cloud Platforms:** AWS, Microsoft Azure
- **Cybersecurity & Systems:** Kali Linux, Nmap, Linux CLI, File Permissions

Soft Skills

- Team Collaboration
- Problem Solving
- Analytical Thinking
- Communication
- Time Management

Education

Adichunchanagiri Institute of Technology

Dec. 2022 – Apr. 2026

Bachelor of Engineering in Computer Science